ORIGINAL ARTICLE
https://doi.org/10.30773/pi.2023.0410

Print ISSN 1738-3684 / On-line ISSN 1976-3026
OPEN ACCESS

Challenges of Positive Airway Pressure Treatment for Patients With Obstructive Sleep Apnea at a Veterans Medical Center

Young Kyung Moon[1], Jaejong Lee[1], and Hayun Choi[1,2] ✉

[1]Department of Psychiatry, Veterans Health Service Medical Center, Seoul, Republic of Korea
[2]Department of Psychiatry, Seoul National University College of Medicine, Seoul, Republic of Korea

Objective Positive airway pressure (PAP) compliance is important in treating obstructive sleep apnea. Previous studies have suggested that patients with economic burdens, discomfort using machines, and insufficient education have difficulty tolerating PAP machines. This study explored the factors affecting short-term adherence to PAP in a veterans medical center.

Methods The medical records of patients who underwent polysomnography at the Veterans Health Service Medical Center, Seoul, Republic of Korea, between July 2018 and January 2021 were reviewed retrospectively. Patients with an apnea-hypopnea index (AHI) ≥15 were included (n=579). PAP adherence was defined as continuous use for ≥21 days for ≥4 hours daily for 30 consecutive days for 90 days from the date of PAP prescription.

Results The PAP-adherent group (n=265, age 66.16±11.28 years) was younger and had more years of education, higher body mass indices, and lower scores in the Insomnia Severity Index and Beck Depression Inventory-II (BDI-II) than those of the PAP-nonadherent group (n=314, age 68.93±10.91 years). Patients who tolerated PAP had a higher AHI, longer duration of oxygen desaturation (less than 90%), and less dream enactment behavior (DEB) than that in those who did not. After adjusting for age, years of education, BDI-II, duration of oxygen desaturation, and presence of DEB, there were more patients with National Health Insurance (NHI) in the PAP-adherent group than in the PAP-nonadherent group (p<0.001).

Conclusion We showed that patients with NHI recorded significantly higher adherence compared to that in patients without NHI, among other factors.

Psychiatry Investig 2024;21(7):701-709

Keywords Obstructive sleep apnea; Polysomnography; Insurance; Veterans.

INTRODUCTION

Sleep apnea is a condition in which breathing temporarily stops or decreases during sleep. This condition is divided into obstructive, central, and mixed types, with obstructive sleep apnea (OSA) being the most common type, accounting for over 80% of cases.[1] OSA is characterized by repetitive periods of breathing cessation lasting for ≥10 sec despite repeated efforts to breathe.[2] Apneas and hypopneas are terminated by temporary awakenings, leading to excessive daytime sleepiness and fatigue. OSA is often reported by patients who experience choking or gasping sensations during sleep or by their bed partners who witness snoring and apnea episodes.[3] The prevalence of OSA in South Korea is reported to be 4.5% in males and 3.2% in females;[4] however, the actual prevalence is estimated to be approximately 20%−30% in middle-aged individuals, as many cases are undiagnosed due to unawareness of the symptoms.[4] OSA is considered an independent risk factor for various chronic diseases, including metabolic disorders[5] and cardiovascular conditions such as hypertension, congestive heart failure, coronary artery disease, and stroke.[6] Additionally, OSA causes neurological and mood disorders such as depression and cognitive impairments.[7,8]

In cases of suspected OSA due to clinical symptoms and signs, a polysomnography study is necessary for the final diagnosis.[9] The treatment of choice for moderate-to-severe OSA involves the use of positive airway pressure (PAP) devices. The PAP device delivers continuous air pressure to the airway to keep it open and promote unobstructed breathing. Numerous studies have suggested that successful treatment is possi-

Received: November 22, 2023 **Revised:** March 17, 2024
Accepted: March 24, 2024

✉ **Correspondence:** Hayun Choi, MD
Department of Psychiatry, Veterans Health Service Medical Center, 53 Jinhwangdo-ro 61-gil, Gangdong-gu, Seoul 05368, Republic of Korea
Tel: +82-2-2225-1158, **Fax:** +82-2-2225-3947, **E-mail:** chhy0402@gmail.com

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ble in >80% of patients with OSA;[9] however, long-term compliance can be a hurdle. When compliance is defined as using the device for a minimum of 4 hours a day, at least 5 days a week, international studies have reported that 30%−80% of patients were compliant. Studies in South Korea before National Health Insurance (NHI) coverage was implemented showed that approximately 40% of patients were compliant.[10-12] After insurance coverage, compliance was 84.8% over 3 months after coverage.[13]

Low compliance can result from factors including frequent awakenings in patients with OSA and insomnia due to the PAP mask preventing sleep, machine noise disrupting sleep, and bulky device size. Other factors contributing to low compliance include skin damage, inflammation, pain caused by wearing a mask, and anxiety or discomfort associated with device usage.[14] In the past, PAP therapy was hindered by high costs; however, since July 2018, it has been covered by the NHI for patients with OSA who were diagnosed using polysomnography, making it more accessible. In South Korea, insurance coverage for PAP treatment is provided and maintained based on patient compliance. Compliance is assessed every 3 months after the initial period, and if the patient does not submit usage records or if the average usage is less than 2 hours a day, insurance coverage is discontinued.[13]

In Korea's clinical setting, several studies have been conducted to investigate changes in device compliance among patients with OSA following the introduction of NHI coverage for PAP devices. Some of these studies reported an increase in PAP compliance at the third- and sixth-month follow-ups, while others found higher compliance rates in the first and third months but no statistically significant differences in the ninth month.[10,13] Additionally, a study reported that even after initiating insurance coverage, 87.9% of patients discontinued PAP therapy within 13 months, and 51.7% of those who discontinued it before 13 months cited device maladaptation as the reason.[15] These findings suggest that while insurance coverage may impact the initial and short-term compliance increase with PAP treatment, other factors need to be considered for long-term compliance, and identifying factors that hinder PAP device usage early on is crucial.

In South Korea, residents are required to enroll in the NHI system and pay insurance premiums. Among them, beneficiaries of medical aid are individuals with income below a certain level who receive medical services provided by the NHI without paying premiums. As an exception, national merit recipients do not enroll in the NHI but instead receive medical services from veterans' hospitals and contracted hospitals (Figure 1).[16]

The Veterans Health Service Medical Center (VHS Medical Center) provides an environment conducive to studying the relationship between NHI coverage and PAP compliance because it has a significant number of patients who are not members of the NHI and receive medical benefits in comparison with those in university hospitals. This study aimed to evaluate short-term PAP adherence in patients who underwent polysomnography and were diagnosed with OSA with an apnea-hypopnea index (AHI) score >15 at the VHS Medical Center and to identify NHI enrollment or medical aid and other related factors affecting PAP adherence.

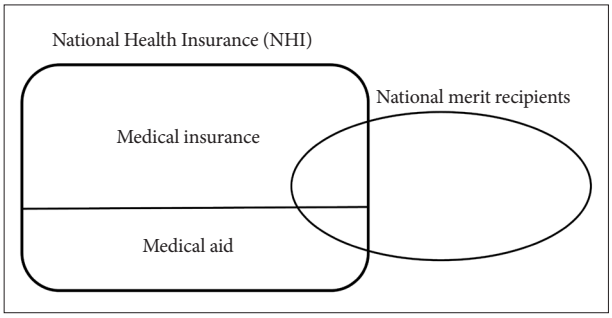

Figure 1. A brief overview of South Korea's National Health Insurance and national merit recipients.

METHODS

Study population

This study recruited 917 patients who underwent nocturnal polysomnography due to symptoms of sleep apnea and were diagnosed with OSA with an AHI >15 between July 1, 2018 and January 31, 2021 at the department of psychiatry of the VHS Medical Center (Figure 2). For the 4 patients who underwent polysomnography twice, the first set of data was selected, and the second set of data was excluded. The participants were divided into two groups as follows: those who adhered to PAP therapy (n=272) and those who did not (n=336). The PAP-adherent group included patients who received their initial PAP prescription and used the device for a minimum of 4 hours a day for more than 21 days out of 30 consecutive days within the first 90 days of receiving the prescription. The nonadherent group included patients who did not meet these criteria. Patients with schizophrenia, delusional disorders, delirium, dementia, or any psychiatric disorder due to medical conditions, as well as those with a history of neurological abnormalities such as head trauma or epilepsy and a PAP buyer, were excluded from the study. Data from 314 patients nonadherent to PAP therapy and 265 patients adherent to PAP therapy were analyzed.

The NHI medical insurance subscribers were 508 among the recruited patients, and they had to pay 34 US dollars for the continuous positive airway pressure (CPAP) machine rental. There were 29 medical aid beneficiaries, and they didn't

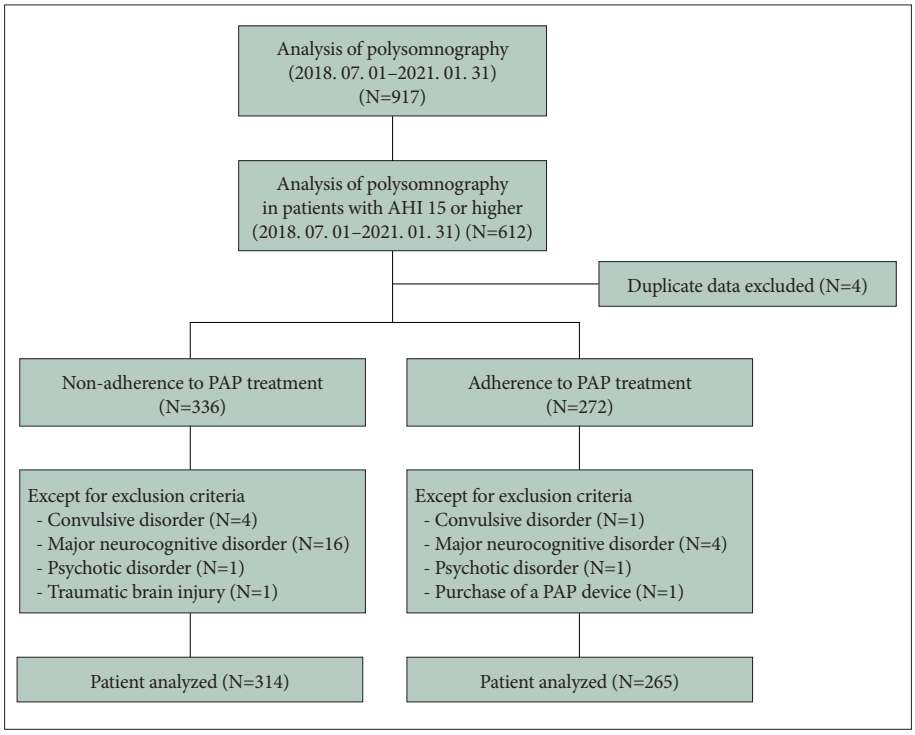

Figure 2. Schematic overview of our patient population. AHI, apnea-hypopnea index; PAP, positive airway pressure.

have to pay the rental fee. The non-NHI subscribers numbered 42, and they had to pay 68 US dollars for the CPAP machine rental.

This retrospective study relied on a medical record review; therefore, the requirement for obtaining informed consent was waived, and the study was approved by the Clinical Trials Review Committee of the VHS Medical Center (IRB No. 2021-02-012).

Clinical information and neuropsychological evaluation

All study participants visited the outpatient clinic of the department of psychiatry and completed self-reported questionnaires before undergoing nocturnal polysomnography. The questionnaires included basic personal information and clinical data, such as age, sex, education, weight, height, body mass index (BMI), sleep-related behaviors, medical history, current medication use, and sleep habits. Clinical psychological assessments were conducted to evaluate daytime sleepiness, subjective sleep quality, and the severity of depressive symptoms. These assessments included the Pittsburgh Sleep Quality Index (PSQI),[17] Insomnia Severity Index (ISI),[18] Epworth Sleepiness Scale (ESS),[19] Korean version of the Dysfunctional Beliefs and Attitudes about Sleep Questionnaire-16,[20] and Beck Depression Inventory-II (BDI-II).[21] Coexisting diseases included sleep disorders such as insomnia, rapid eye movement (REM) sleep behavior disorder (RBD), periodic limb movement disorder (PLMD), psychiatric disorders such as depression and anxiety disorders, and internal medical conditions including cardiovascular and respiratory diseases. Diagnoses of sleep disorders, such as RBD and PLMD, were determined by reviewing medical records and nocturnal polysomnography results. To confirm depression, medical records were reviewed, and BDI-II and the use of antidepressants, benzodiazepines, and Z-drugs were considered. The presence of internal medical conditions, such as cardiovascular and respiratory diseases, was determined through a review of medical records and patient medical histories.

Nocturnal polysomnography

The sleep study was conducted using the Comet-PLUS PSG (Grass Instrument Co., Warwick, RI, USA), and standardized electrodes and sensors were used. Electrodes for brainwave measurements were attached at F4/A1, F3/A2, C4/A1, C3/A2, O1/A2, and O2/A1, following the international 10−20 electrode placement system. In addition, two ocular muscle sensors were affixed to the lateral sides of both eyes to record the horizontal and vertical eye movements. To monitor lower-limb movements during sleep, electromyography (EMG) sensors were attached to both anterior tibialis muscles, and an EMG sensor was placed above the submentalis muscle to track chin muscle activity. Strain gauges were used to detect chest and abdominal respiratory movements, whereas nasal pressure cannulas were used to measure nasal airflow. Blood oxy-

gen saturation (SaO_2<90) was measured using a sensor attached to the tip of the index finger of the left hand. Sleep architecture was analyzed based on criteria established in accordance with standardized guidelines published by the American Academy of Sleep Medicine.[2]

Statistical analysis

To identify differences between the PAP-adherent sleep apnea group and the nonadherent group in demographic characteristics, coexisting diseases, polysomnography results, sleep-related questionnaire variables, and PAP usage patterns, independent t-tests were used for continuous variables and chi-squared tests were used for categorical variables in the association analysis.

To investigate the impact of NHI-related factors on PAP adherence, a univariate logistic regression analysis was performed, including confounding factors. Statistically significant variables were selected from the univariate analysis, and multiple logistic regression analysis was performed using backward stepwise selection. We identified several confounding factors using independent t-tests and chi-square tests. The confounding factors included age, years of education, BMI, ISI score, BDI-II, SaO_2<90, spontaneous arousal, and dream enactment behavior (DEB). All tests were two-tailed, and p-values <0.05 were considered statistically significant. SPSS (version 16.0; SPSS Inc., Chicago, IL, USA) and R version 3.6.3 (R Foundation for Statistical Computing, Vienna, Austria; http://www.r-project.org) were used as the software programs for statistical analyses.

RESULTS

Sociodemographic characteristics

The total number of participants in this study was 579, with 265 in the PAP-adherent group and 314 in the nonadherent group (Figure 2). The PAP-adherent group had a mean age of 66.16±11.28 years, which was significantly lower than that of the nonadherent group (68.93±10.91 years, p=0.003). Additionally, the PAP-adherent group had a higher level of education (p=0.001) and a higher BMI (p=0.012) (Table 1). Neuropsychological evaluation results showed that the PAP-adherent group exhibited a higher ESS (p=0.023), lower ISI (p=0.041), and lower BDI-II (p=0.012). However, no significant difference was observed in the total PSQI score, although the sleep efficiency component of the PSQI was higher and wake time after sleep onset (WASO) was shorter in the adherent group. Among the two groups, there was no significant difference in the prevalence of comorbid internal medicine conditions, such as diabetes, cardiovascular disease and cerebrovascular disease. Diagnoses of insomnia (p=0.001), use of antidepressants (p=0.026), and benzodiazepine-type drugs (p<0.001) were lower in the PAP-adherent group; however, the frequency of Z-drug use did not show statistical significance. Enrollment in NHI was also associated with PAP adherence (p<0.001).

Nocturnal polysomnography and PAP data

The polysomnography results showed that the PAP-adherent group exhibited higher values for certain parameters than those in the nonadherent group. These parameters included a higher AHI (p=0.018), longer duration of SaO_2<90 (p=0.004), longer duration of arousal (p=0.007), and lower frequency of DEB (p=0.014). However, there were no significant differences between the two groups in terms of parameters such as total sleep time, WASO, sleep efficiency, sleep duration, REM sleep duration, PLMD, mouth breathing, or REM sleep without atonia (Table 2).

Analysis of factors affecting PAP compliance

The results of the multinomial logistic regression analysis comparing factors affecting PAP adherence showed that patients who were not enrolled in the NHI exhibited significantly lower adherence than that of those with health insurance (p< 0.001). Among medical aid beneficiaries, there was no significant difference in adherence compared with that in patients with medical insurance. Additionally, younger age (p=0.026), more years of education (p=0.019), longer duration of SaO_2<90 (p=0.002), and lower DEB (p=0.031) were associated with higher adherence (Table 3).

DISCUSSION

This study showed that patients who were enrolled in the NHI recorded significantly higher 3-month PAP adherence than that in patients without NHI after considering factors such as insomnia, depression, age, years of education, duration of oxygen desaturation, presence of DEB, and BMI. Moreover, patients who benefited from medical aid did not have a higher 3-month PAP adherence than that in patients with medical insurance.

The 3-month PAP adherence rate among the study participants was 45.8%. This result is significantly lower than the 84.8% adherence rate reported in another 3-month study conducted at a tertiary hospital.[13] Before the NHI-covered PAP treatment, one institution reported a 6-month adherence rate of 35.7%[22] and another reported a 41.5% adherence rate,[11] however, this was based on a different time frame than the 3-month period in this study, making direct comparisons challenging. Several factors can influence PAP therapy adherence, including patient-, physician-, and device-related fac-

Table 1. Demographics and clinical characteristics of eligible participants

	Non-adherence to PAP treatment (N=314)	Adherence to PAP treatment (N=265)	p
Age (yr)	68.93±10.91	66.16±11.28	0.003*
Sex			0.724
Men	296 (94.27)	247 (93.21)	
Women	18 (5.73)	18 (6.79)	
Education (yr)	12.01±3.63	12.95±3.20	0.001*
BMI (kg/m^2)	25.51±3.48	26.23±3.44	0.012*
PSQI			
Total	8.64±4.25	7.98±3.76	0.056
TIB (hour)	7.35±1.48	7.20±1.31	0.214
TST (hour)	6.04±1.58	6.26±1.42	0.091
Sleep efficiency (%)	82.88±16.61	87.27±14.61	0.001*
Sleep latency (min)	33.93±31.17	30.16±31.06	0.156
WASO (hour)	0.75±1.29	0.44±1.10	0.003*
ESS	9.13±5.82	10.24±5.63	0.023*
ISI	11.35±7.19	10.17±6.32	0.041*
DBAS-16	89.20±33.53	89.60±31.84	0.886
BDI-II	14.33±12.72	12.00±8.54	0.012*
Comorbidity			
Diabetes mellitus	106 (33.76)	77 (29.06)	0.262
Angina	52 (16.56)	33 (12.45)	0.203
Myocardial infarction	34 (10.83)	16 (6.04)	0.058
Cerebrovascular disease	31 (9.87)	31 (11.70)	0.567
Insomnia	103 (32.80)	53 (20.00)	0.001*
Medication			
Antidepressants	103 (32.80)	64 (24.15)	0.026*
Benzodiazepines	129 (41.08)	48 (18.11)	<0.001*
Z-drugs	18 (5.73)	11 (4.15)	0.498
NHI			<0.001†
Medical insurance	257 (81.8)	251 (94.72)	
Medical aid beneficiary	19 (6.1)	10 (3.77)	
No insurance	38 (12.1)	4 (1.51)	

Data are presented as mean±SD or number (%). *p<0.05; †vs. no insurance group, p<0.001. PAP, positive airway pressure; BMI, body mass index; PSQI, Pittsburgh Sleep Quality Index; TIB, time in bed; TST, total sleep time; WASO, wake time after sleep onset; ESS, Epworth Sleepiness Scale; ISI, Insomnia Severity Index; DBAS-16, Korean version–Dysfunctional Beliefs and Attitudes about Sleep Questionnaire-16; BDI-II, Beck Depression Inventory-II; NHI, National Health Insurance; SD, standard deviation

tors.[23] This study's more detailed examination focused on cost burden on the patient, age, and psychological factors. Specifically, some potential reasons for the relatively low PAP adherence observed in this study include the following: a unique patient population of the hospital where the research was conducted, as it primarily serves national merit recipients and their families; a high proportion of older patients; the operation of a sleep clinic within the department of psychiatry that may have influenced the patient population and their adherence; and the high prevalence of trauma histories in patients that led to the observation of DEB.

The cost of renting a PAP device may be a barrier or facilitator of adherence. Before the NHI started covering PAP treatment in South Korea, monthly rental fees varied by the company but were generally approximately 200,000−300,000 KRW (approximately 150−225 US dollars). After NHI coverage, monthly rental fees were approximately 34 US dollars before the adherence period and 13−15 US dollars after. As of June

2018, research findings suggest that this change in rental fees led to increased PAP adherence.[13]

However, the VHS Medical Center, where this study was conducted, primarily serves national merit recipients, such as veterans and their family members, and has a population of patients who are not enrolled in the NHI. This factor has been shown to have a significant impact on PAP adherence. South Korea has implemented the NHI system for the entire nation, covering nearly 99% of the population. However, national merit recipients such as veterans and their family members have the option to choose whether to enroll in the NHI or opt out according to Article 5, Paragraph 1 of the NHI Act.[16] In-

Table 2. Polysomnographic findings of eligible participants

	Non-adherence to PAP treatment (N=314)	Adherence to PAP treatment (N=265)	p
TST (min)	285.52±63.28	292.72±57.42	0.155
WASO (min)	95.71±60.70	90.49±53.08	0.270
Sleep latency (min)	20.71±28.93	17.86±24.34	0.198
Sleep efficiency (%)	71.39±15.13	73.32±13.87	0.113
Stage 1 (%)	21.99±13.86	22.30±14.67	0.796
Stage 2 (%)	58.09±12.77	56.67±11.48	0.164
Stage 3 (%)	1.84±5.35	1.93±4.77	0.824
REM (%)	18.26±9.88	19.44±10.37	0.163
AHI	37.56±19.43	45.91±54.39	0.018*
NRAHI	38.29±20.45	43.10±21.95	0.007*
RAHI	30.64±22.44	38.59±22.04	<0.001*
SaO_2<90 (%)	7.89±12.70	11.11±13.99	0.004*
$MinSaO_2$ (%)	81.39±7.01	79.68±7.02	0.004*
PLMI	15.57±25.62	13.41±25.64	0.312
PLM arousal	2.39±4.91	2.33±7.01	0.913
Mouth breathing	13 (4.14)	16 (6.04)	0.394
RSWA	63 (20.06)	44 (16.60)	0.336
DEB	34 (10.83)	13 (4.91)	0.014*

Data are presented as mean±SD or number (%). *p<0.05. PAP, positive airway pressure; TST, total sleep time; WASO, wake time after sleep onset; REM, rapid eye movement; AHI, apnea-hypopnea index; NRAHI, non rapid eye movement apnea-hypopnea index; RAHI, rapid eye movement apnea-hypopnea index; SaO_2, oxygen saturation; PLMI, periodic limb movement index; PLM arousal, periodic limb movements with arousal; RSWA, rapid eye movement sleep without atonia; DEB, Dream enactment behavior; SD, standard deviation

Table 3. Multinomial logistic regression analyses

Outcome	Predictor	Simple		Multiple	
		OR (95% Cl)	p	OR (95% Cl)	p
Compliance of PAP	Group not enrolled in NHI*	0.11 (0.04−0.31)	<0.001	0.10 (0.03−0.28)	<0.001
	Medical aid beneficiary*	0.54 (0.25−1.18)	0.123	0.57 (0.25−1.32)	0.188
	Age (yr)	0.98 (0.96−0.99)	0.003	0.98 (0.96−1.00)	0.026
	Education (yr)	1.08 (1.03−1.14)	0.002	1.07 (1.01−1.13)	0.019
	SaO_2<90 (%)	1.02 (1.01−1.03)	0.005	1.02 (1.01−1.04)	0.002
	DEB	0.42 (0.22−0.82)	0.011	0.44 (0.21−0.93)	0.031
	BMI (kg/m^2)	1.06 (1.01−1.12)	0.013		
	ISI	0.97 (0.95−1.00)	0.043		
	BDI-II	0.98 (0.96−1.00)	0.015	0.98 (0.97−1.00)	0.073
	AHI after PAP	1.02 (0.90−1.15)	0.783		

*using group enrolled in NHI as reference. OR, odds ratio; CI, confidence interval; PAP, positive airway pressure; NHI, National Health Insurance; SaO_2, oxygen saturation; DEB, dream enactment behavior; BMI, body mass index; ISI, Insomnia Severity Index; BDI-II, Beck Depression Inventory-II; AHI, apnea-hypopnea index

dividuals who opt out of the NHI or do not have health insurance can receive medical benefits at the VHS Medical Center and affiliated hospitals. In such cases, family members can benefit from the reduced medical fees when seeking treatment at the VHS Medical Center. In this study, 42 out of 579 patients (7%) were not enrolled in the NHI, which differs from the 1% rate on not-enrollment in the NHI among the entire population. Consequently, the high number of patients without an NHI at VHS Medical Center has led to different rates of PAP device application based on health insurance status. Even among health insurance enrollees, many patients may not pay any or most of their medical expenses, as they qualify for reduced or waived fees. This is particularly relevant for PAP device rental, where some patients may find the burden of costs unfair. In the clinical setting, it has been observed that some patients become upset when a doctor recommends PAP device rental after undergoing polysomnography because they are required to bear the cost. Some patients may expect to rent the device for free, leading them to consider changing hospitals to pursue such benefits.

Meanwhile, as a public hospital, the VHS Medical Center often receives medical aid beneficiary patients regardless of whether they are national merit recipients or not. Patients benefiting from medical aid can use PAP devices without incurring personal costs. Economically, patients receiving medical aid are expected to have significantly higher PAP adherence. However, the results of this study showed that the presence of medical aid did not have a statistically significant effect on PAP adherence. It was challenging to emphasize that NHI support simply increases compliance because compliance did not increase for medical aid beneficiaries who can rent the machine for free. For patients receiving medical aid, it seems like economic factors do not act as luring factor since they don't need to pay personal cost regardless of adherence. This suggests that not only economic factors, but also various factors, including age, years of education, insomnia, and depression, may collectively influence adherence rates.

In this study, the average age of both groups was >65 years, indicating an older population compared to that in a previous study.[13] Individuals aged ≥65 years are often retired from stable employment and may rely on low or no work-related income, depending on pension income for their livelihood. This demographic group may experience economic concerns related to the rental cost of PAP devices.

In this study, the group meeting the PAP adherence criteria was younger than the nonadherent group, suggesting that age can impact PAP compliance. Previous research from the USA reported varying adherence rates based on age groups; the percentage of individuals meeting the PAP adherence criteria increased significantly from 54.7% in the age group of 18−30 years to 79.0% in the 61−70 years age group and then slightly decreased to 73.1% in the 81−90 years age group.[24] These results show that the age group with the highest PAP adherence rate was not necessarily the youngest, indicating that PAP adherence is influenced by multiple factors beyond age. In a similar study conducted in South Korea, the "good adherence" group tended to be younger than the "poor adherence" group; however, statistically significant differences were not observed.[13] These results regarding adherence vary according to age and sex, which could be influenced by differences in each country's insurance and older adults' welfare systems or cultural norms.[25]

Furthermore, as this study was conducted on patients visiting a sleep clinic within the department of psychiatry, there may be a higher distribution of individuals with insomnia than that in studies conducted in another department, such as otolaryngology. Individuals who failed to meet the PAP adherence criteria had a higher prevalence of insomnia and higher BDI-II scores than those of individuals who met the adherence criteria. This likely reflects the significant impact of a patient's low-arousal threshold[26] and hyperarousal state on PAP adherence.[27] A low arousal threshold in patients with OSA refers to the tendency to wake up easily in response to a relatively mild airway obstruction during sleep. This condition is one of the physiological characteristics that occurs in over one-third of patients with OSA and can result in difficulty maintaining sleep, leading to insomnia.[26] Additionally, individuals with insomnia and depression may exhibit overactivation of the arousal system, possibly due to psychophysiological hyperarousal caused by increased orexin, which plays a role in both the arousal and affective systems.[27]

In this study, when comparing the results of polysomnography between the PAP-adherent and nonadherent groups, the PAP-adherent group had a higher AHI, a longer duration of SaO_2<90, and a lower occurrence of DEB. Previous studies have shown no strong association between PAP adherence and AHI severity or nocturnal hypoxemia.[13,23] However, another study showed that good adherence groups tended to have higher AHI and significantly higher non-rapid eye movement sleep and supine AHI.[13] The need for PAP may be emphasized in patients with higher AHI severity or nocturnal hypoxemia. The observation that individuals with DEB were less likely to achieve good adherence could be related to the presence of idiopathic REM RBDs and trauma-related sleep disorders.[28] Some patients may exhibit body movements or vocalizations during sleep due to conditions such as combat trauma or military trauma-related sleep disorders, making the use of PAP therapy challenging.

Furthermore, the study found that individuals who achieved PAP adherence had more years of education than those who

did not. This suggests that a patient's understanding of the necessity of PAP therapy may influence adherence. Doctors often need to explain the importance of PAP therapy to patients with OSA within a limited time frame, and some patients may find it difficult to comprehend. Therefore, research has suggested that both face-to-face and e-health interventions can extend the duration of PAP use, emphasizing the importance of patient education and support in improving adherence.[29]

This study had some limitations. First, due to the characteristics of the VHS Medical Center where the study was conducted, there was an unequal sex distribution (93.8% males) and a high proportion of patients over the age of 65 years, making it difficult to represent the entire patient population. Second, the study defined the PAP-adherent group as individuals who used PAP therapy for a minimum of 4 hours a day for more than 21 days out of 30 consecutive days within the first 90 days of receiving the prescription, which allows for an assessment of short-term adherence only. As previous research has shown that adherence can change depending on the duration of follow-up, future studies should consider analyzing the factors affecting long-term adherence based on the results of this study. Also, it was challenging to find prior research in South Korea comparing compliance rates between medical aid and medical insurance. This difficulty may stem from the rarity of medical aid beneficiaries visiting hospitals engaged in research and article publication, making it hard to secure research participants. In addition, to assess psychiatric conditions, the study used information such as BDI-II and the use of antidepressant medications; hence, the actual diagnosis of mood or anxiety disorders was not confirmed. Furthermore, because the study was retrospective, we could only rely on information recorded by the authors. Finally, it is worth noting that the study did not consider findings related to structural abnormalities of the nose, such as nasal septum deviation, or factors such as the Mallampati score, that reflect the upper airway structure. Additionally, medical conditions such as rhinitis or asthma, which can potentially influence PAP adherence, were not considered in this study. These factors can also affect the patient's ability to adhere to PAP therapy and could be important for a more comprehensive understanding of the topic.

Despite these limitations, this study is valuable in its ability to identify patient characteristics associated with difficulties in PAP adherence. It would be good for clinicians to consider these factors and use them as a reference to increase compliance.

Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Hayun Choi. Data curation: Hayun Choi, Jaejong Lee. Formal analysis: Hayun Choi, Jaejong Lee. Funding acquisition: Hayun Choi. Investigation: Hayun Choi. Methodology: Hayun Choi. Project Administration: Young Kyung Moon, Hayun Choi. Resources: Hayun Choi. Software: Hayun Choi. Supervision: Hayun Choi. Validation: Young Kyung Moon. Visualization: Hayun Choi. Writing—original draft: Young Kyung Moon, Jaejong Lee. Writing—review and editing: Young Kyung Moon, Hayun Choi.

ORCID iDs

Young Kyung Moon https://orcid.org/0000-0003-0797-5516
Jaejong Lee https://orcid.org/0009-0003-6819-9327
Hayun Choi https://orcid.org/0000-0002-8404-8006

Funding Statement

This study was supported by a grant from the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare, Republic of Korea (Grant No. HI21C0852).

Acknowledgments

We express our deepest gratitude to Young Lee (Veterans Medical Research Institute) and Aram Park (Veterans Health Service Medical Center), who made valuable contributions to the statistical analyses.

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